

TTK4215 Adaptive Control
Fall 2026

Instructor: Professor Ole Morten Aamo, Office: D344, email: aamo@ntnu.no

Assistant: Jon Erik Karsrud Nordal, email: jenordal@stud.ntnu.no

Time and Location:

Lectures: Monday 12-15 EL1

Exercises: Thursday 8-10 H1 (Hovedbygget)

Textbook: Ioannou & Fidan, Adaptive Control Tutorial, Society for Industrial and Applied Mathematics 2006.

Material (pensum): Entire I&F book, lecture notes and exercises.

Course content: The course gives an introduction to methods for adaptation and learning in control of dynamic systems with uncertain parameters. On-line parameter estimation: gradient methods and least squares methods in continuous and discrete time. Parameter estimation with projection. Extremum seeking methods. Direct and indirect adaptive control: pole placement control (PPC), adaptive pole placement control (APPC), model reference control (MRC), model reference adaptive control (MRAC), adaptive backstepping with tuning functions.

Grading: There will be 12 assignments to be completed independently, and which will be given pass/no pass grading. 5 out of the 12 assignments must be completed with passing grade to be eligible for the final exam. The final exam is the basis for a letter grade in the course.

Final exam (tentative): December 19th, 09:00-13:00, (D - calculator). Digital on Inspera.

Tentative Lecture plan

Week	Topic	
34	I&F 1.3, A.3, 3.2, 4.1.	
35	I&F 3.3-4, 2.	Assignment 1 due
36	I&F 3.6.	Assignment 2 due
37	I&F 3.7.	Assignment 3 due
38	I&F 3.10-12.	Assignment 4 due
39	I&F 4.	Assignment 5 due
40	Extremum seeking	Assignment 6 due
41	I&F 5.1-2.	Assignment 7 due
42	I&F 5.3.	Assignment 8 due
43	I&F 6.3-6.5.	Assignment 9 due
44	I&F 7.2-7.3.	Assignment 10 due
45	Adaptive control of nonlinear systems	Assignment 11 due
46	Adaptive control of nonlinear systems	Assignment 12 due
47	NO LECTURE: MAKEUP WEEK FOR COMPLETING ASSIGNMENTS	DUE: 2 makeup assignments

Assignments are posted on Canvas along with deadlines (Fridays at 23:59). Papers must be delivered electronically on Canvas and should consist of one single pdf file (scanned handwriting is acceptable).